

Module 5

Two-Port Networks and Network Graphs

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Hybrid parameters (h-parameters)

The z and y parameters of a two-port network do not always exist. So there is a need for developing another set of parameters. This third set of parameters is based on making V_1 and I_2 the dependent variables.

$$\begin{aligned} V_1 &= h_{11}I_1 + h_{12}V_2 \\ I_2 &= h_{21}I_1 + h_{22}V_2 \end{aligned}$$

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix} = [\mathbf{h}] \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$

The h_{11} , h_{12} , h_{21} , h_{22} are known as the **hybrid parameters** (or, simply, h parameters) because they are a hybrid combination of ratios.

$$\begin{aligned} h_{11} &= \left. \frac{V_1}{I_1} \right|_{V_2=0}, & h_{12} &= \left. \frac{V_1}{V_2} \right|_{I_1=0} \\ h_{21} &= \left. \frac{I_2}{I_1} \right|_{V_2=0}, & h_{22} &= \left. \frac{I_2}{V_2} \right|_{I_1=0} \end{aligned}$$

h_{11} = Short-circuit input impedance

h_{12} = Open-circuit reverse voltage gain

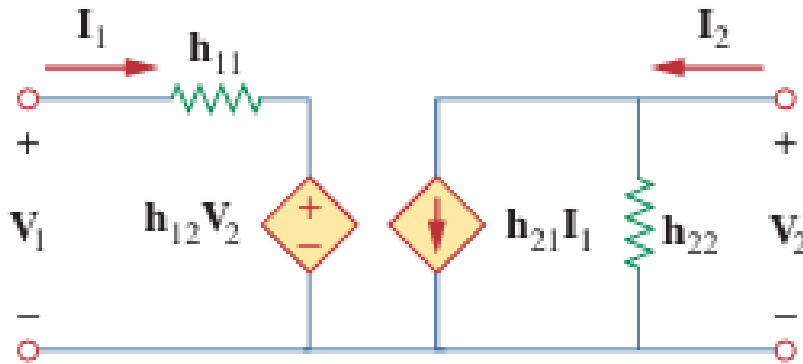
h_{21} = Short-circuit forward current gain

h_{22} = Open-circuit output admittance

For reciprocal networks, $h_{12} = -h_{21}$

h- parameters

Hybrid model of a two-port network:



A set of parameters closely related to the h parameters are the g parameters or **inverse hybrid** parameters. These are used to describe the terminal currents and voltages as:

$$\begin{aligned} I_1 &= g_{11}V_1 + g_{12}I_2 \\ V_2 &= g_{21}V_1 + g_{22}I_2 \end{aligned}$$

$$\begin{bmatrix} I_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = [g] \begin{bmatrix} V_1 \\ I_2 \end{bmatrix}$$

The h parameters are frequently used to model Bipolar junction transistors.

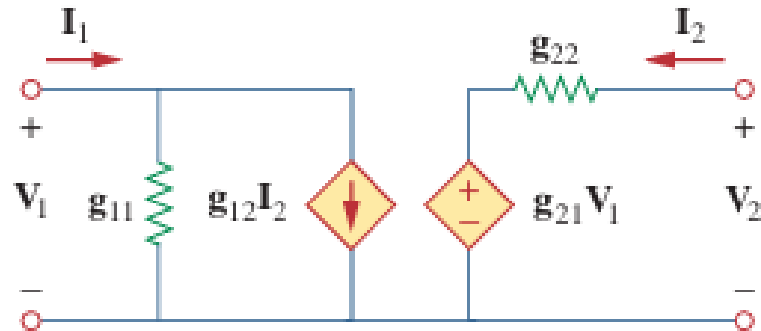
g- parameters

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$$\begin{aligned} g_{11} &= \left. \frac{I_1}{V_1} \right|_{I_2=0}, & g_{12} &= \left. \frac{I_1}{I_2} \right|_{V_1=0} \\ g_{21} &= \left. \frac{V_2}{V_1} \right|_{I_2=0}, & g_{22} &= \left. \frac{V_2}{I_2} \right|_{V_1=0} \end{aligned}$$

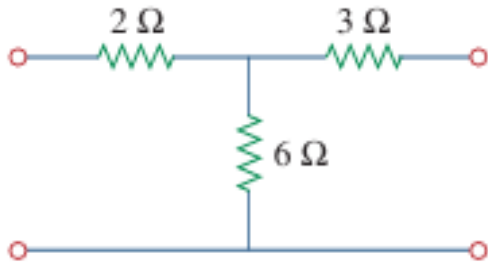


- g_{11} = Open-circuit input admittance
- g_{12} = Short-circuit reverse current gain
- g_{21} = Open-circuit forward voltage gain
- g_{22} = Short-circuit output impedance

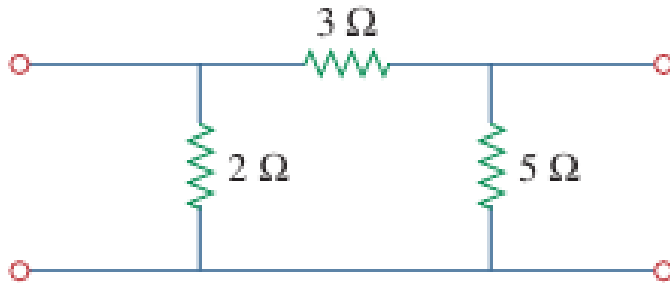
The g parameters are frequently used to model field-effect transistors.

h - parameters

1. Find the hybrid parameters for the two-port network.

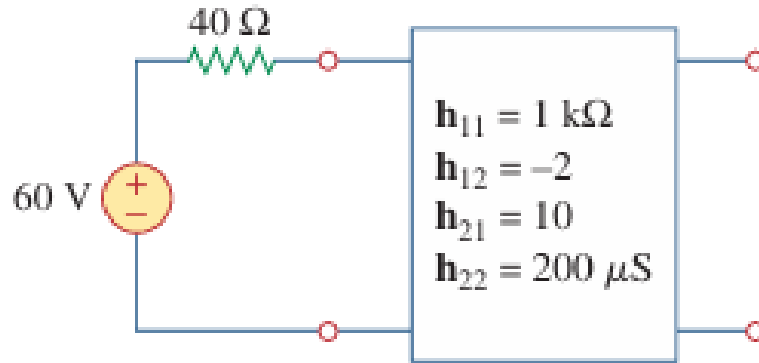


2 Determine the h parameters for the circuit.

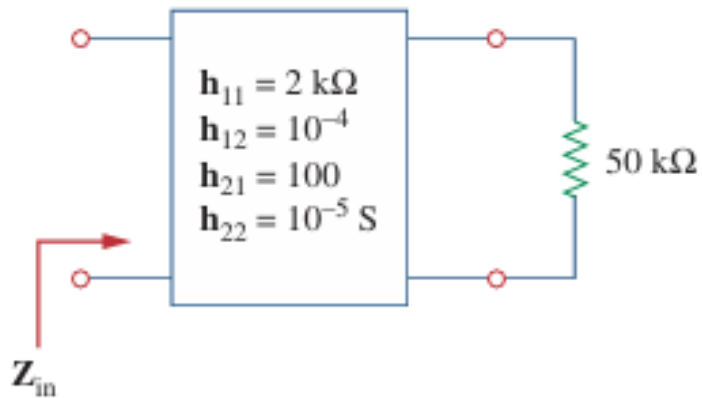


h - parameters

3. Determine the Thevenin equivalent at the output port of the circuit.

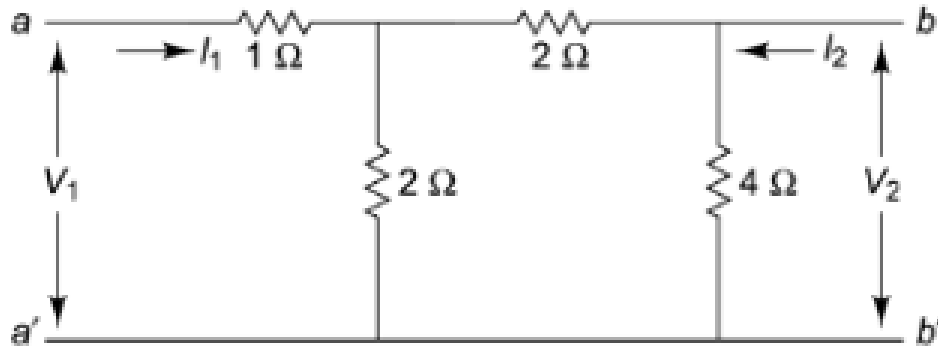


4. Find the impedance at the input port of the circuit.

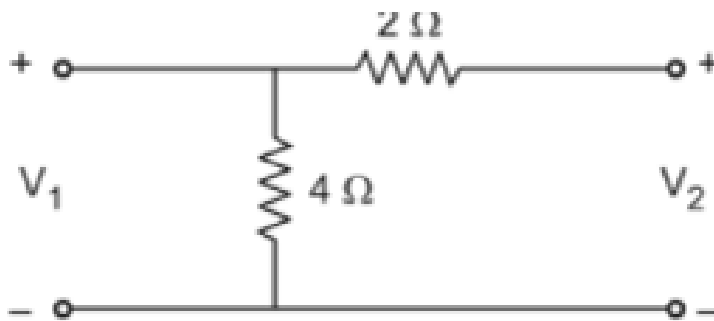


h - parameters

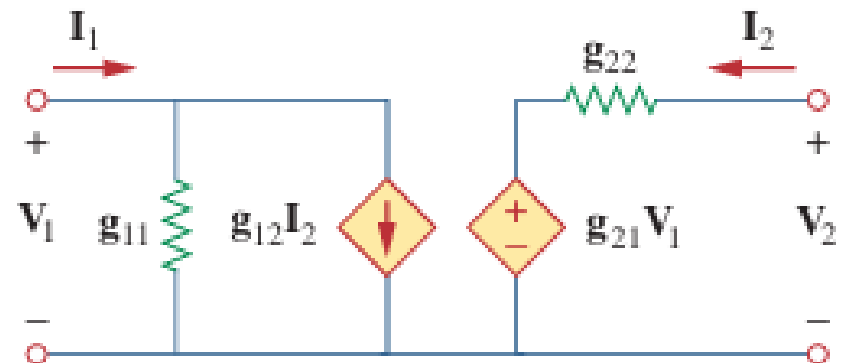
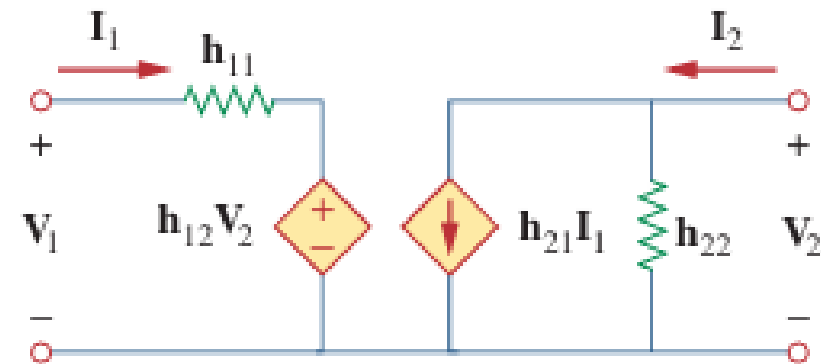
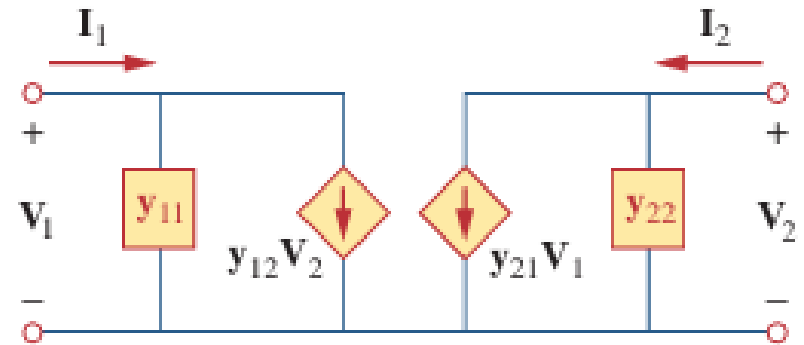
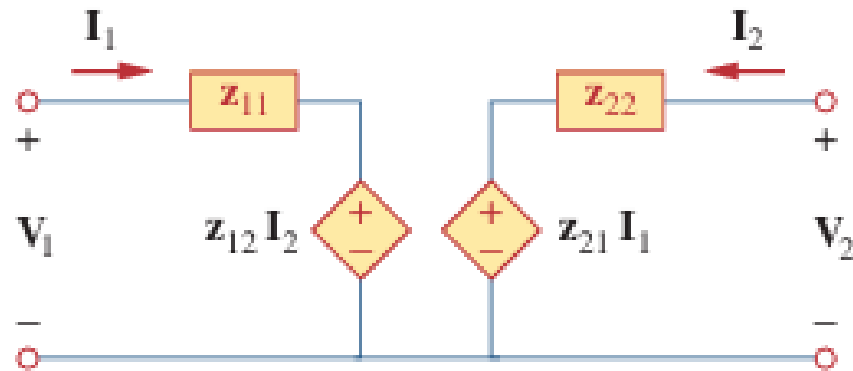
5. Find the h parameters of the network shown.



6. Find the hybrid parameters for the two-port network.



Equivalent circuits with circuit parameters



Next class: Transmission line parameters